Politecnico di Milano
School of Industrial and Information Engineering
Bayesian Learning and Monte Carlo Simulation

Final project:
Yacht Hydrodynamic Resistance

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Contents

1 Data and Problem

We analyse the *Yacht Hydrodynamics* dataset from the UCI Machine Learning Repository [1, 2]: 22 sailing-yacht hull forms tested at several speeds, for $n = 308$ observations. Each row holds five hull *form factors* — `LCB`, `prismatic`, `L_disp`, `beam_draught` and `L_beam` (constant within a hull) — and the dimensionless **Froude number** $Fn = v/\sqrt{gL} \in [0.125, 0.45]$; the response is the residuary resistance per unit weight of displacement (Table 4).

For slender hulls, physics predicts a power law resistance $\propto Fn^{\gamma_F}$ with $\gamma_F \approx 4$. We study it in a Bayesian framework:

1. log–log regression to estimate and *verify* $\hat{\beta}_F \approx 4$, with a prior-sensitivity check;
2. BAS/BMA over the 2^5 hull-factor submodels, to see which carry signal once Fn is included;
3. a heteroscedasticity-corrected nonlinear model on the retained factors, contrasted with the log–log fit;
4. posterior predictive curves, with and without the heteroscedasticity treatment, against the data.

2 Log–log Regression

2.1 Model and prior

Taking logarithms turns the physical power law resistance $\propto Fn^{\gamma_F}$ into a linear model in which the exponent appears directly as a regression slope. We therefore fit

$$\log(\text{resistance}) = \beta_0 + \beta_F \, \log(\text{Froude}) + \sum_{k=1}^{5} \beta_k \, \text{hull}_k + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

where hull_k denotes the five geometric form factors; the smallest observed resistance is $0.01 > 0$, so the logarithm is well defined for every record and no offset is required. As a baseline we adopt the non-informative **Jeffreys reference prior**

$$\pi(\boldsymbol{\beta} \mid \sigma^2) \ \propto \ 1, \qquad \pi(\sigma^2) \ \propto \ \sigma^{-2}, \quad (2)$$

which lets the data dominate the inference; the sensitivity of the conclusions to an informative prior is examined in section 2.4.

2.2 Posterior and credible intervals

With the Jeffreys reference prior in Eq. (2), the Gaussian linear model has a closed-form posterior. Let $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$. Then

$$\boldsymbol{\beta} \mid \sigma^2, \boldsymbol{y} \sim \mathcal{N}\Big(\hat{\boldsymbol{\beta}}, \, \sigma^2 \, (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\Big), \quad (3)$$

Integrating out σ^2 gives a multivariate Student-t posterior for $\boldsymbol{\beta}$ with $n - p$ degrees of freedom, so the 95% credible intervals coincide numerically with the classical OLS confidence intervals.

2.3 Results

Table 5 (in the appendix) reports the posterior summaries. The model explains the data very well ($R^2 = 0.97$, residual standard deviation $s = 0.32$ on the log scale) and the posterior of the Froude exponent is sharply concentrated,

$$\beta_F \approx 4.69, \qquad 95\% \text{ CI} = [4.60,\ 4.78],$$

confirming the expected power-law behaviour resistance $\sim Fn^4$ and verifying the project requirement $\hat{\beta}_F \approx 4$.

Fig. 3 (appendix) shows the log–log scatter with the posterior-mean line and its very tight 95% credible band (left), and the posterior of β_F (right): the whole posterior mass sits above the nominal value 4 ($\Pr(\beta_F > 4 \mid \boldsymbol{y}) \approx 1$), so the empirical exponent is best described as *slightly steeper* than the textbook value of four.

2.4 Sensitivity check: independent informative priors

The reference-prior analysis *verifies* $\beta_F \approx 4$ a posteriori; a complementary check *encodes* the physics in the prior and asks how strongly the prior must insist on 4 before it overrides the data. We replace Eq. (2) with **independent proper priors**,

$$\beta_F \sim \mathcal{N}(4, \tau^2), \qquad \beta_j \sim \mathcal{N}(0, 100^2) \ \ (j \neq F), \qquad \sigma^2 \sim \text{Inv-Gamma}(10^{-3}, 10^{-3}), \tag{4}$$

all components mutually independent. Since $\boldsymbol{\beta}$ and σ^2 are now independent a priori, unlike in the conjugate Normal–Inverse-Gamma family, the joint posterior is no longer available in closed form, but both full conditionals remain standard,

$$\boldsymbol{\beta} \mid \sigma^2, \boldsymbol{y} \sim \mathcal{N}\big(\boldsymbol{V}(\boldsymbol{S}_0^{-1}\boldsymbol{m}_0 + \boldsymbol{X}^\top \boldsymbol{y}/\sigma^2),\ \boldsymbol{V}\big), \qquad \boldsymbol{V} = (\boldsymbol{S}_0^{-1} + \boldsymbol{X}^\top \boldsymbol{X}/\sigma^2)^{-1}, \tag{5}$$

$$\sigma^2 \mid \boldsymbol{\beta}, \boldsymbol{y} \sim \text{Inv-Gamma}\big(a_0 + \tfrac{n}{2},\ b_0 + \tfrac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2\big), \tag{6}$$

where $\boldsymbol{m}_0$ and $\boldsymbol{S}_0$ collect the prior means and variances of Eq. (4). We therefore sample the posterior with a **Gibbs sampler** (10 000 iterations, 2 000 burn-in; the effective sample size for β_F exceeds 4 000 in every run) and sweep the prior strength $\tau \in \{0.05, 0.1, 0.2, 0.5\}$.

Table 6 (appendix) summarises the sweep; the posterior densities are overlaid in Fig. 6 (appendix). A moderately informative prior ($\tau = 0.5$) reproduces the reference result. Tightening the prior pulls the posterior mean towards 4, but only the near-dogmatic $\tau = 0.05$ (prior 95% interval $\approx [3.9, 4.1]$) drags it as far as 4.33, and even then $\Pr(\beta_F > 4 \mid \boldsymbol{y}) = 1$.

For any moderately diffuse physical prior ($\tau \geq 0.1$) the likelihood dominates and the posterior mean stays in 4.56–4.68. The conclusion that the empirical exponent is genuinely steeper than Fn^4 is therefore **robust to the prior**: it is a feature of the $n = 308$ observations, not an artefact of the non-informative choice, a transparent instance of prior–data conflict, in which an informative likelihood overwhelms all but an unreasonably confident prior.

3 BAS/BMA model selection over hull covariates

3.1 Model space and forced Froude effect

The log–log regression in section 2 establishes that the dominant physical scaling is the Froude power law. The model-selection question is therefore conditional on Froude: after log(Froude) has been included, do any hull-geometry variables still carry independent information about log(resistance)? For this reason the term log(Froude) is forced into every candidate model; the BAS/BMA step is not intended to test whether speed matters, but to identify possible secondary hull effects after speed has been accounted for.

The optional covariates are the five hull form factors `LCB`, `prismatic`, `L_disp`, `beam_draught` and `L_beam`. They are treated as optional because the full log–log regression does not resolve any of them individually: their credible intervals cross zero, and the form factors are strongly related across the limited set of 22 hull designs. With five optional variables, the model space contains only $2^5 = 32$ submodels. We can therefore enumerate the whole space with `bas.lm`, using `include.always = ~ log_Fr` and a uniform prior over the 32 hull-covariate submodels.

This averaging approach is preferable to interpreting only the full regression coefficients. The full model conditions on all five hull covariates being present simultaneously, even though they are correlated and weakly identified. BAS/BMA instead averages over all possible subsets and reports, for each covariate, its posterior inclusion probability (PIP): the posterior probability that the covariate belongs in the regression model, conditional on log(Froude) being included.

3.2 Baseline BAS/BMA results

The baseline analysis uses the BIC marginal-likelihood approximation in `bas.lm` with a uniform model prior. The resulting PIPs are shown in Table 1. A large PIP would indicate that a covariate appears repeatedly in high-posterior-probability models, while a small PIP means that the data do not require that term once Froude and the competing hull variables are considered.

Table 1: Approximate posterior inclusion probabilities for the five optional hull-geometry covariates when log(Froude) is forced into every model. Values are based on enumeration of the $2^5 = 32$ submodels using the BIC marginal-likelihood approximation, matching the baseline `prior = "BIC"` specification in `bas.lm`.

Covariate	PIP	Interpretation
`beam_draught`	0.57	moderate, not decisive
`prismatic`	0.27	weak
`LCB`	0.21	weak
`L_disp`	0.21	weak
`L_beam`	0.19	weak

The largest PIP is for `beam_draught`, approximately 0.57. This is moderate evidence, not decisive evidence: the variable is the most plausible secondary hull covariate under the baseline prior, but it is far from being overwhelmingly selected.

The other four hull variables have PIPs well below 0.50, so their evidence is weak conditional on log(Froude). This agrees with the full-model coefficient intervals in Table 5, but is more defensible because it checks the whole model space rather than relying on a single saturated regression.

3.3 Sensitivity to prior choices

Because the baseline PIPs are modest, the BAS/BMA conclusion was checked under alternative BAS/BMA prior specifications. The same model space was enumerated in each run: log(Froude) was forced into every model, the five hull covariates were optional, and the model prior over the $2^5 = 32$ submodels was kept uniform. The prior specifications compared were the baseline `BIC / reference` approximation, Zellner's `g-prior (g = n)`, and the heavy-tailed `JZS` prior. The resulting PIPs are reported in Table 2.

Prior specification	LCB	prismatic	L_disp	beam_draught	L_beam
`BIC / reference`	0.210	0.265	0.206	0.565	0.192
`g-prior (g = n)`	0.186	0.217	0.182	0.499	0.166
`JZS`	0.060	0.057	0.051	0.243	0.042

Table 2: Sensitivity of posterior inclusion probabilities to alternative BAS/BMA prior specifications.

The sensitivity analysis reinforces the need for caution. The robust conclusion is not that `beam_draught` is definitely selected. In fact, no hull covariate reaches PIP ≥ 0.50 under all three prior specifications, and the magnitude of the inclusion probabilities changes noticeably. The robust conclusion is instead that all hull covariates are much weaker than the forced Froude effect once log(Froude) is included. At the same time, `beam_draught` remains the best-supported secondary covariate in the baseline analysis and the top-ranked hull covariate in each sensitivity run, so it remains a reasonable candidate rather than an arbitrary choice. The observed prior sensitivity motivates a cautious interpretation; it does not invalidate the analysis, but it prevents any strong claim that a hull covariate has been unambiguously selected.

3.4 Interpretation and link with the nonlinear model

Overall, BAS/BMA suggests that `beam_draught` is the most defensible secondary covariate, while the Froude power law remains the dominant robust signal. This gives a coherent reduced modelling strategy for the next section: the heteroscedastic nonlinear analysis should compare a Froude-only model M_0 with a model M_1 that also includes `beam_draught`. In this workflow, BAS/BMA proposes the candidate hull covariate, and the nonlinear original-scale model checks whether that covariate improves posterior predictive performance once the variance growth with resistance is modelled explicitly.

4 Heteroscedasticity-Corrected Nonlinear Model

4.1 Nonlinear model

Following the BAS/BMA analysis, the only hull-form variable carried forward was the beam-draught ratio with posterior inclusion probability was about 0.565, so the evidence was moderate rather than strong. Therefore, we fitted both a Froude-only model **M0** and a model including beam-draught **M1**, but the main heteroscedastic nonlinear model was

$$Y_i \mid \theta, \sigma_0, \delta \sim N(\mu_i, \sigma_i^2),$$

where Y_i is the observed residuary resistance and

$$\mu_i = k\,\mathrm{Froude}_i^{\gamma_F} \exp\{\beta_B\,\mathrm{bd}_i\}.$$

where bd_i is the beam–draught ratio (`beam_draught`), and $k > 0$ is the multiplicative constant of the power law. To model heteroscedasticity, we did not assume a constant variance. Instead, we used

$$\sigma_i = \sigma_0 \mu_i^{\delta}.$$

The parameter δ controls how fast the standard deviation grows with the fitted mean. If $\delta = 0$, then the model reduces to a homoscedastic additive-error model on the original scale. If $\delta = 1$, then the standard deviation is proportional to the mean, which is close to the multiplicative-error interpretation of the log-log model.

The parameters to be estimated are the constant k, the Froude exponent γ_F, the hull-coefficient β_B, the baseline noise level σ_0, and the **variance-growth exponent** δ. We have include the test of heteroscedasticity correction by comparing the fixed version of the variance ($\delta = 0$) and version with learned variance with WAIC and PSIS-LOO tests.

For computation, the mean was rewritten in exponential form:

$$\mu_i = \exp\{\log k + \gamma_F \log(\mathrm{Froude}_i) + \beta_B \mathrm{bd}_i\}.$$

In the implementation we centered log(Froude) and beam-draught before fitting the model. Writing $\log \mathrm{Froude}_i = m_F + \tilde{x}_{F,i}$ and $\mathrm{bd}_i = m_B + \tilde{x}_{B,i}$ and absorbing the constants into a single intercept b_0,

$$b_0 = \log k + \gamma_F m_F + \beta_B m_B, \tag{7}$$

$$\mu_i = \exp\big\{b_0 + \gamma_F\,\tilde{x}_{F,i} + \beta_B\,\tilde{x}_{B,i}\big\}. \tag{8}$$

Centring relabels the intercept but leaves the slopes γ_F and β_B untouched, so it does not change the inference on the quantities of physical interest. After sampling, the physical constant is recovered draw by draw from

$$k = \exp\big\{b_0 - \gamma_F m_F - \beta_B m_B\big\}. \tag{9}$$

4.2 Prior distributions

All priors are weakly informative and mutually independent. For the mean parameters,

$$b_0 \sim \mathcal{N}(0, 10^2), \qquad \gamma_F \sim \mathcal{N}(4, 2^2), \qquad \beta_B \sim \mathcal{N}(0, 2^2). \tag{10}$$

The prior on γ_F is centred at the physical value 4. For the variance function,

$$\log \sigma_0 \sim \mathcal{N}(0, 10^2), \qquad \delta \sim \mathcal{N}(0, 1^2). \tag{11}$$

The prior on δ is centred at 0, i.e. at homoscedasticity, precisely so that any evidence of variance growth comes from the data and not from the prior. As a sensitivity check on the headline quantity, the γ_F prior is also replaced by a vague $\mathcal{N}(0, 10^2)$ and by an informative $\mathcal{N}(4, 0.5^2)$.

4.3 Posterior sampling algorithm

Unlike log-log linear model, the posterior distribution is not availiable in closed form. The likelihood contains a nonlinear exponential mean, and the variance also depends on the mean through $\sigma_i = \sigma_0 \mu_i^{\delta}$. Because of this non-conjugacy, we used MCMC in JAGS.

For this model JAGS automatically assigned the univariate slice sampler to the continuous parameters.

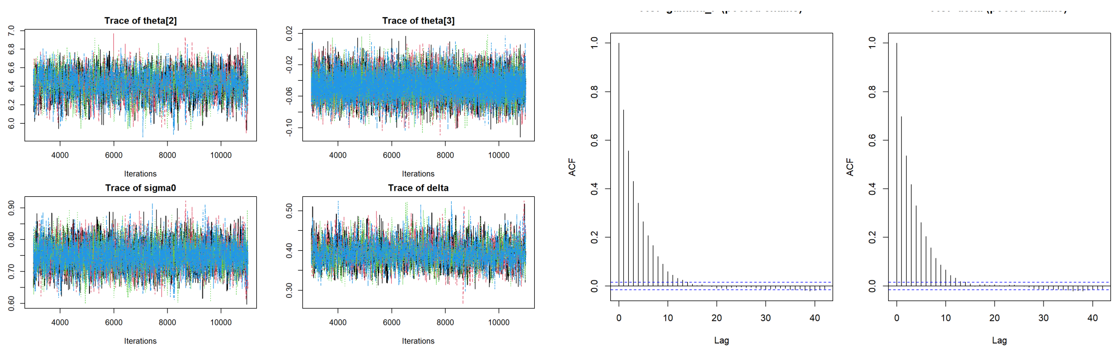
We run four independent chains, started from dispersed values of γ_F ($3, 6, 5, 4$) and of δ (values $0, 1, 0.5, 1.2$) Each chain uses an adaptation phase, a burn-in that is discarded, and a thinned sampling phase. If chains launched from very different points all settle on the same distribution, that is good evidence the sampler has reached the posterior rather than a starting-value-dependent region.

4.4 Convergence diagnostics

Before doing any intrepretion from posterior distribution or sampling from posterior, we should check wheter our proposed model has converged. We checked the converges of the non-linear with several different methods. First we check Gelman–Rubin statistic $\hat{R}$. This compares the variance between the dispersed chains to the variance within each chain. If the chains have all converged to the same distribution the two variances agree. Secondly, we check Effective sample size (ESS). Successive MCMC draws are autocorrelated, so N kept draws carry less information than N independent ones. The ESS estimates the equivalent number of independent draws; we want it in the thousands so that posterior means and quantiles are estimated with little Monte-Carlo error. Trace plots, for each parameter the four chains should look like stationary noise fluctuating around a common, fixed level, with the chains overlapping rather than drifting or sitting at separate values. Finally, Autocorrelation function (ACF), For each parameter the autocorrelation of the chain should decay to zero within a few lags.

Table 3: Convergence diagnostics for the heteroscedastic nonlinear model M1 under the main prior.

Parameter	$\hat{R}$	Effective sample size
δ	1.0000	2208
σ_0	1.0000	3284
b_0	1.0002	4409
γ_F	0.9999	2182
β_B	1.0003	13012



(a) Trace plots　　**(b)** Posterior of β_F.

Figure 1: Trace plots of the variables, each color represents a chain.

Table 3 reports the convergence diagnostics for the main M1 run. All Gelman–Rubin statistics are essentially equal to one, indicating that the independently initialized chains are sampling from the same posterior distribution. The effective sample sizes are also sufficiently large for posterior inference, especially for β_B. Figure 1 shows that different chains overlap on the plot and look stationary noices, it shows that every chained mixed with the given sampling numbers, and ACF decays to zero in few lags.

4.5 Posterior analysis

4.6 Model comparison

We use two questions to validate the modelling choices, both answered with WAIC and PSIS-LOO computed from the pointwise log-likelihood $\log p(y_i \mid \boldsymbol{\theta}^{(s)})$ of the MCMC draws (lower WAIC/LOOIC is better; PSIS-LOO additionally returns Pareto-$\hat{k}$ diagnostics that warn of overly influential observations).

First question is that was the BAS analysis true?
Second question is was learned variance correct the heteroscedasticity.

5 Posterior Predictive Curves

The final comparison is made on the original resistance scale, because this is where the data are observed and where the two error assumptions imply visibly different uncertainty bands. For a fair comparison, both curves in Fig. 2 are posterior predictive curves: each posterior draw includes uncertainty in the mean curve and one new observation-level error term. The observed yacht-resistance data are overlaid on the same scale.

For the log–log regression, the posterior predictive distribution is lognormal after transforming back to resistance. After drawing $(\boldsymbol{\beta}, \sigma^2)$ from the posterior, replicated values are generated as

$$\log y_i^{\text{rep}} \sim \mathcal{N}(x_i^\top \boldsymbol{\beta}, \sigma^2), \qquad y_i^{\text{rep}} = \exp(\log y_i^{\text{rep}}).$$

For the heteroscedastic nonlinear M1 model, replicated values are generated on the original scale with a mean-dependent standard deviation,

$$y_i^{\text{rep}} \sim \mathcal{N}(\mu_i, \sigma_i^2), \qquad \mu_i = \exp\{b_0 + \gamma_F \widetilde{\log Fn}_i + \beta_B \widetilde{bd}_i\}, \qquad \sigma_i = \sigma_0 \mu_i^{\delta}.$$

The displayed curves hold the hull form fixed at a representative value: `beam_draught` is set to its sample median, while the other log–log hull covariates are set to their sample means. This isolates the Froude–resistance relationship instead of mixing different hull geometries.

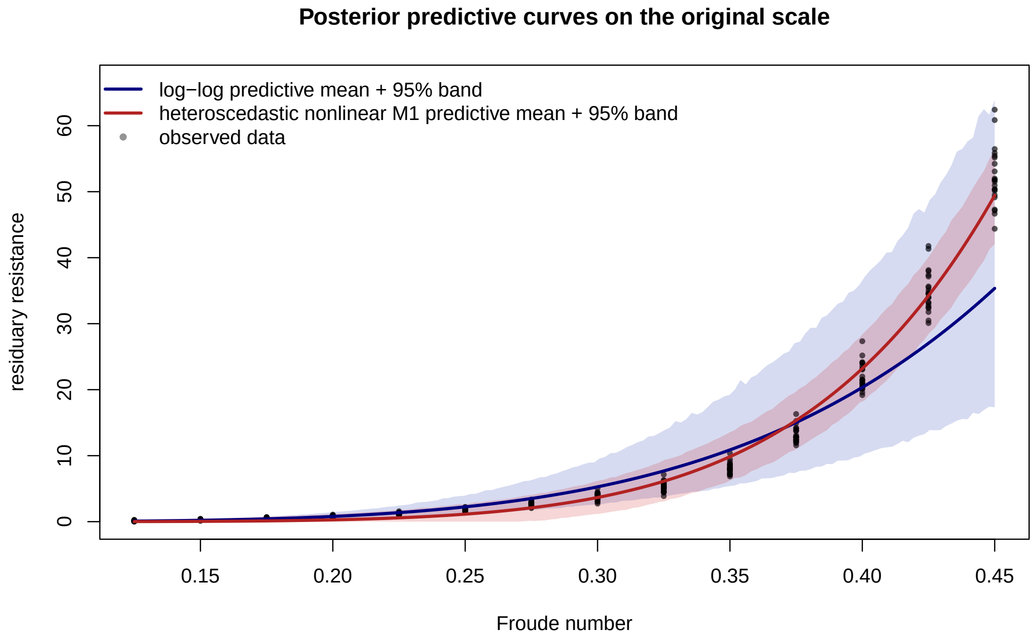


Figure 2: Posterior predictive curves on the original resistance scale. The blue band is the log–log model after back-transformation, and the red band is the heteroscedastic nonlinear M1 model. Points are the observed yacht resistance data.

The log–log fit keeps the Froude exponent close to the physical fourth-power law and produces multiplicative uncertainty. Its band is tight near zero resistance and expands as resistance grows, which describes the low and middle Froude range reasonably well. However, the mean curve is too flat at the highest Froude values, where the observed resistance rises faster than the fitted $Fn^{4.69}$ relationship.

The heteroscedastic nonlinear fit behaves differently. Its mean curve is steeper ($\gamma_F \approx 6.4$) and follows the high-Froude observations more closely. Because its variance is learned through $\sigma_i = \sigma_0 \mu_i^{\delta}$ with $\delta \approx 0.4$, the predictive band is not forced to have constant width. It widens where resistance and scatter are larger and tightens toward the low-Froude region. This is the practical payoff of the heteroscedasticity treatment: the uncertainty band adapts to the data rather than imposing a single residual scale across the whole Froude range.

Overall, the posterior predictive curves show that the two models differ mainly in how they scale uncertainty with resistance. The log–log model gives a useful positive predictive distribution, but it imposes multiplicative uncertainty and misses part of the high-speed curvature. The heteroscedastic nonlinear model learns the variance growth directly, tracks the observed data more closely, and supports the conclusion that the treatment of heteroscedasticity is essential for interpreting both the Froude exponent and the selected hull covariate `beam_draught`.

References

[1] I. Ortigosa, R. Lopez, and J. Garcia. *Yacht Hydrodynamics*. UCI Machine Learning Repository. DOI: 10.24432/C5XG7R. 2007.

[2] J. Gerritsma, R. Onnink, and A. Versluis. *Geometry, Resistance and Stability of the Delft Systematic Yacht Hull Series*. Technical Report. Delft University of Technology, Ship Hydromechanics Laboratory, 1981.

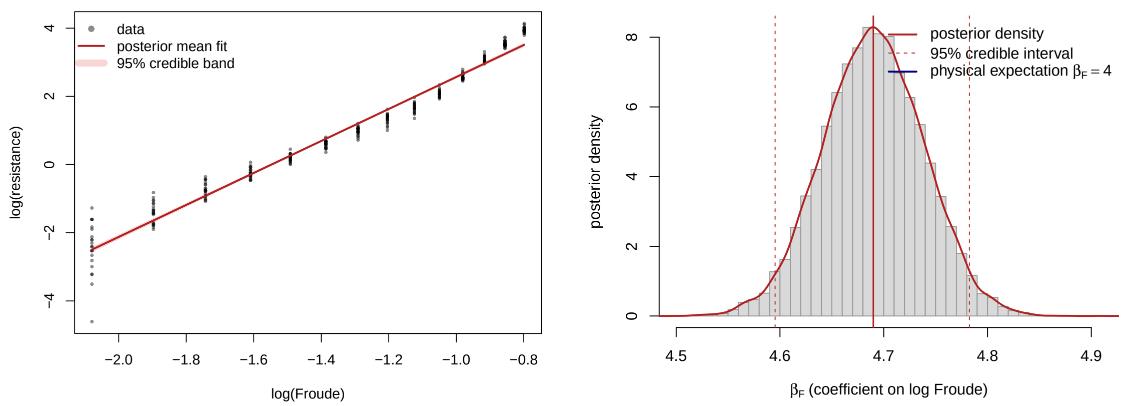
A Tables and Figures

Table 4: Variables of the yacht hydrodynamics dataset.

Variable	Description
LCB	Longitudinal position of the centre of buoyancy
prismatic	Prismatic coefficient
L_disp	Length–displacement ratio
beam_draught	Beam–draught ratio
L_beam	Length–beam ratio
Froude	Froude number, $Fn = v/\sqrt{gL}$
resistance	Residuary resistance per unit weight of displacement

Table 5: Posterior summary of the log–log model in Eq. (1) under the reference prior. Point estimates and 95% credible intervals coincide with the OLS fit (`lm`/`confint`); Pr(> 0) is the posterior probability that the coefficient is positive.

Coefficient	Mean	SD	2.5%	97.5%	Pr(> 0)
Intercept β_0	7.183	0.983	5.248	9.119	1.00
log(Froude) β_F	4.690	0.048	4.596	4.785	1.00
LCB	0.022	0.012	−0.003	0.046	0.96
prismatic	−0.621	1.601	−3.771	2.529	0.35
L_disp	0.465	0.513	−0.546	1.475	0.82
beam_draught	−0.066	0.200	−0.460	0.327	0.37
L_beam	−0.464	0.515	−1.477	0.549	0.18



(a) Power-law fit on log–log axes. (b) Posterior of β_F.

Figure 3: Left: observed log(resistance) against log(Froude) with the posterior-mean regression line (hull covariates held at their means) and its 95% credible band. Right: Student-t posterior density of the Froude exponent β_F; the dashed lines mark the 95% credible interval and the solid blue line the physical reference value $\beta_F = 4$.

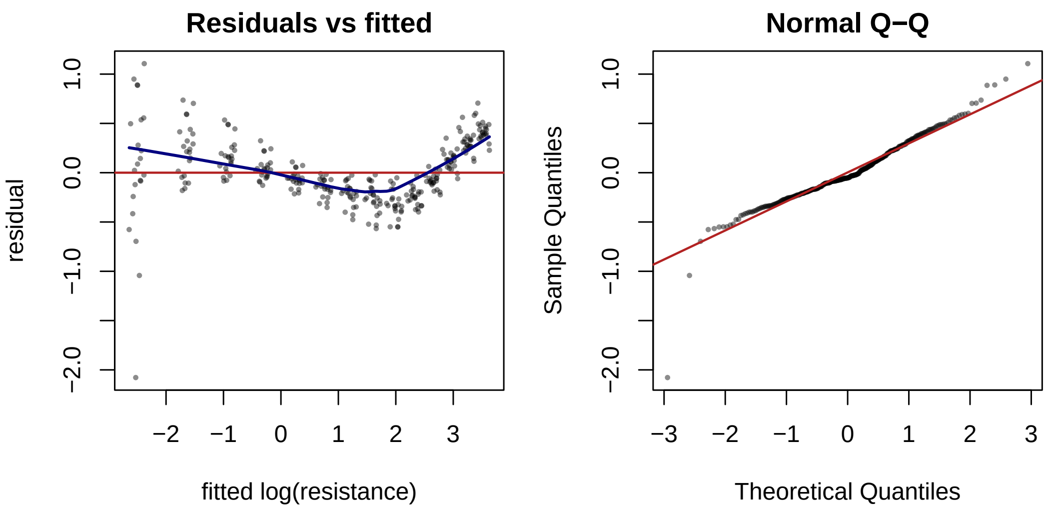


Figure 4: Residual diagnostics for the log–log model. Left: residuals versus fitted values, with a LOWESS trend (blue) revealing speed-dependent curvature and a variance that shrinks as the fitted value grows. Right: Normal Q–Q plot, showing heavier-than-Gaussian lower tails.

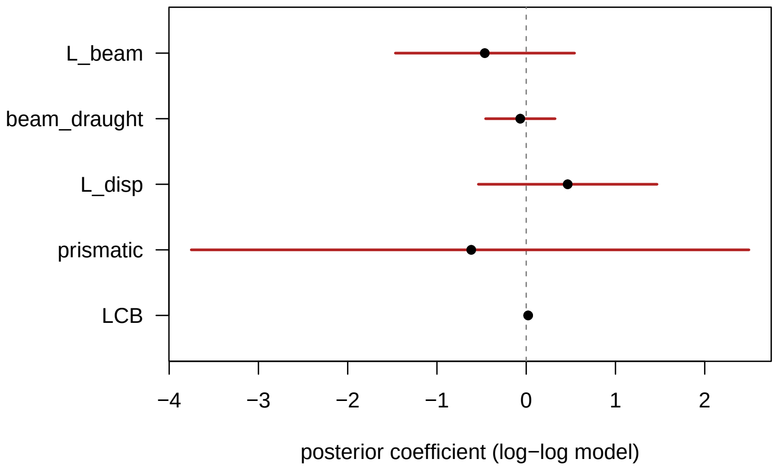


Figure 5: Posterior means and 95% credible intervals of the five hull-geometry coefficients in the log–log model. Every interval straddles zero (dashed line), indicating that no single form factor is resolved once the Froude number is accounted for.

Hull-geometry covariates. Once the Froude number is in the model, none of the five hull form factors is individually credible: every 95% interval comfortably contains zero (Table 5 and Fig. 5). The form factors vary only across the 22 hulls and are strongly collinear — the `prismatic` coefficient, for instance, has a posterior standard deviation of about 1.6. This near-redundancy is precisely what the BAS/BMA model selection in section 3 resolves, through posterior inclusion probabilities rather than single-model p-values.

Residual diagnostics. Although the global fit is excellent, the residuals (Fig. 4) reveal two systematic defects. First, a ∪-shaped trend against the fitted values: a single constant exponent slightly over-predicts at intermediate speeds and under-predicts at the highest speeds. Second, the residual spread is markedly larger at low fitted values, i.e. at low Froude numbers where the resistance is tiny and dominated by measurement noise — a textbook signature of heteroscedasticity, also visible as the fanning of the data cloud on the left of Fig. 3. Both findings directly motivate the heteroscedasticity-corrected nonlinear model developed in the main text.

Table 6: Posterior of the Froude exponent β_F under the independent informative priors in Eq. (4) of increasing strength, obtained by Gibbs sampling, compared with the closed-form reference-prior result.

Prior on β_F	**Mean**	**2.5%**	**97.5%**	$\Pr(\beta_F > 4)$
$\mathcal{N}(4, 0.05^2)$	4.33	4.25	4.41	1.00
$\mathcal{N}(4, 0.10^2)$	4.56	4.47	4.64	1.00
$\mathcal{N}(4, 0.20^2)$	4.65	4.56	4.75	1.00
$\mathcal{N}(4, 0.50^2)$	4.68	4.59	4.78	1.00
reference $\pi(\boldsymbol{\beta}, \sigma^2) \propto \sigma^{-2}$	4.69	4.60	4.78	1.00

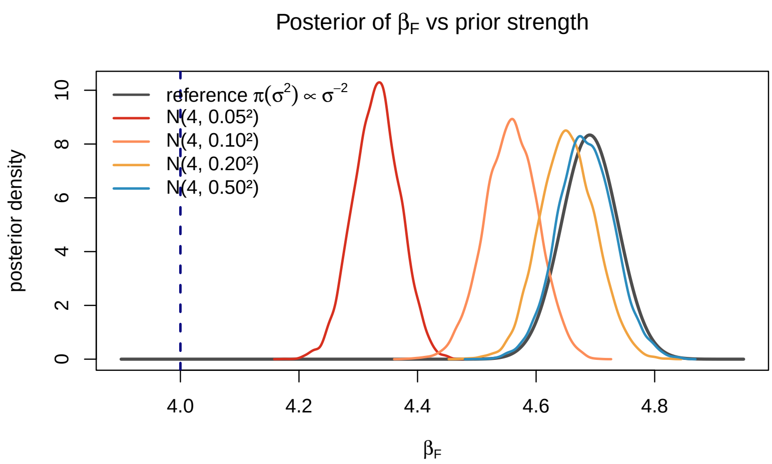


Figure 6: Posterior density of the Froude exponent β_F under the independent informative priors $\mathcal{N}(4, \tau^2)$ of increasing strength (coloured, Gibbs samples) and under the reference prior (grey, closed-form Student-t). The dashed blue line marks the physical value $\beta_F = 4$. Tightening the prior toward 4 shifts the posterior only modestly; the data keep it well above 4 unless the prior is unreasonably confident.